

EES 102: Introduction to Environmental Sciences

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The global natural oxygen cycle

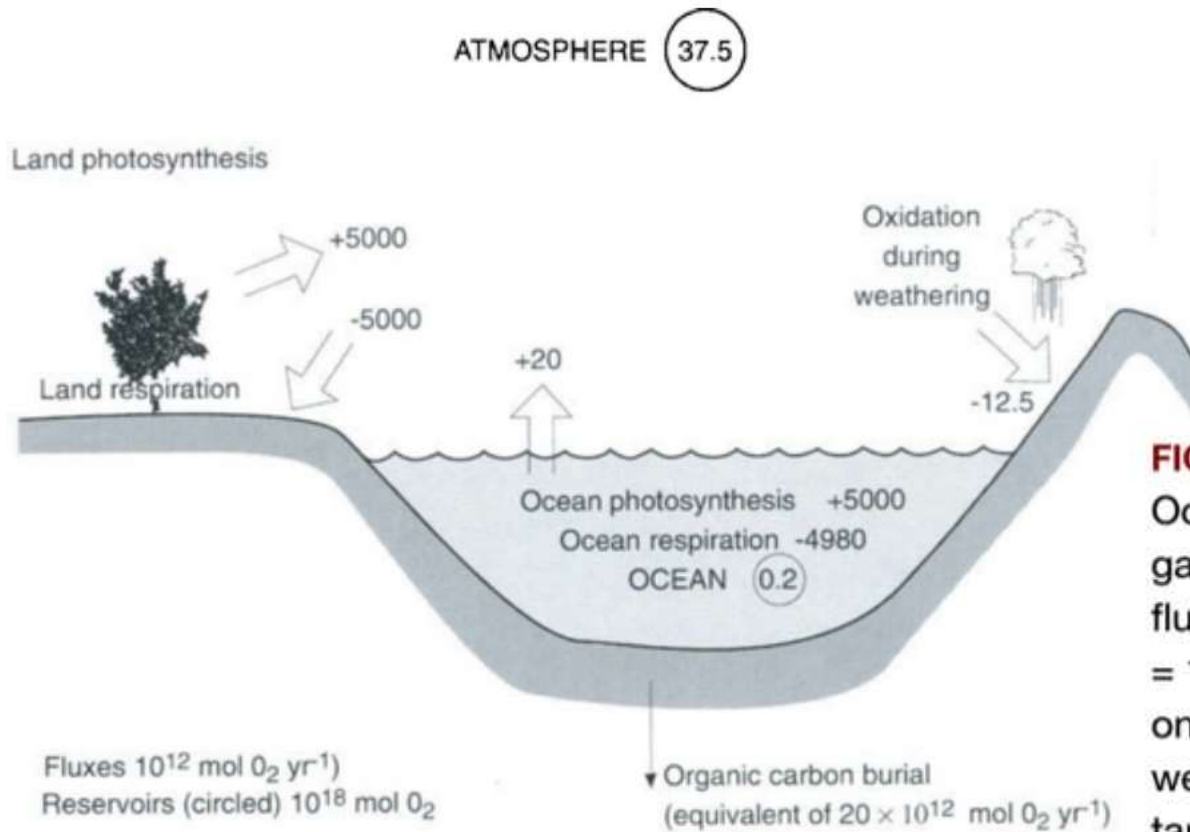


FIG. 16-9 The global oxygen balance. Ocean photosynthesis is estimated. Organic C fluxes are converted to oxygen fluxes using a stoichiometry of $\Delta\text{O}_2/\Delta\text{OC} = 1.5$ in the marine environment and 1.1 on land. Oxygen consumption during weathering of C, S, and Fe in sedimentary rocks, in the ratio of 6:2:1, is. At steady state organic carbon burial and O_2 consumption during weathering must be equal, which is certainly possible within the errors of these two estimates.

from satellite color ([Antoine et al., 1996](#)). It is assumed that photosynthesis on land and in the ocean are about equal. The global carbon burial rate is from [Hedges and Keil \(1995\)](#) from [Holland \(1978\)](#)

In today's atmosphere production of O_2 via photosynthesis and consumption of O_2 via respiration are almost in balance

The global oxygen cycle

The oxygenation of the atmosphere happened when the metabolic pathway of photosynthesis had evolved but the metabolic pathway of respiration had not yet evolved or was not very widespread among biota

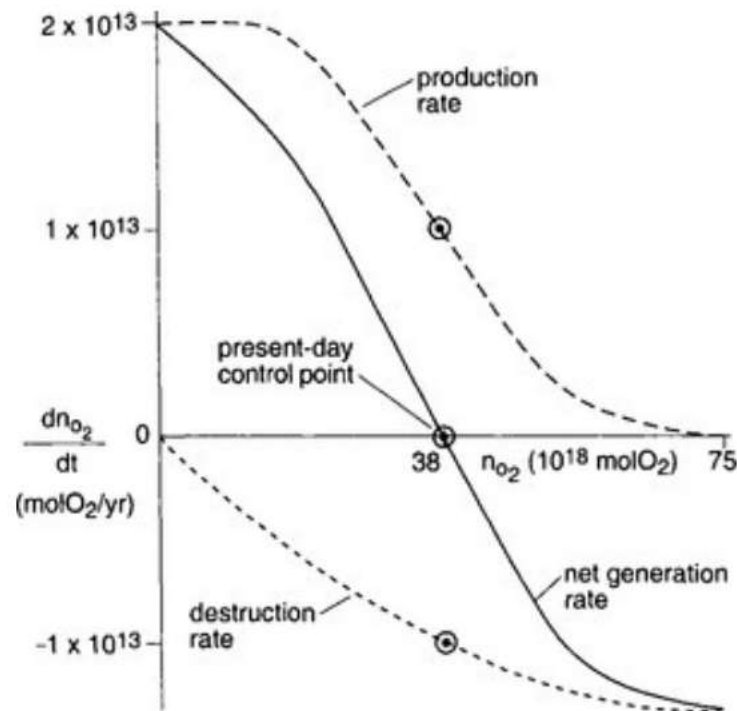


FIG. 16-10 A schematic representation of how the oxygen reduction and production rate, dn_{O_2}/dt combine to determine the size of the global oxygen reservoir. (Adapted from [Holland, 1978](#) .)

THE EON

- The primary and largest catalogued divisions of time within Earth's history are periods called eons.
- The first eon was the Hadean. It started with the formation of the Earth and lasted over 600 million years until the Archean.
- By the **Archean** Earth had cooled enough for **continents and the earliest known life** to emerge. The Archean had a reducing atmosphere and chemoautotrophic organisms
- After about 2.5 billion years, oxygen generated by **photosynthesizing single-celled organisms** began to appear in the atmosphere marking the beginning of the **Proterozoic**. This is the period in which the stratospheric ozone layer formed. The great oxidation event marks the first mass extinction.
- Finally, the **Phanerozoic** eon encompasses 541 million years of **diverse abundance of multicellular life** starting with the appearance of hard animal shells in the fossil record and continuing to the present.

THE ERA

- A geologic era is a subdivision of geologic time that divides an eon into smaller units of time. The Phanerozoic Eon is divided into three such time frames: the Paleozoic, Mesozoic, and Cenozoic (meaning "old life", "middle life" and "recent life") that represent the major stages in the macroscopic fossil record. These eras are separated by catastrophic extinction boundaries: the P-T boundary between the Paleozoic and the Mesozoic, and the K-Pg (also known as K-T) boundary between the Mesozoic and the Cenozoic

PERIOD

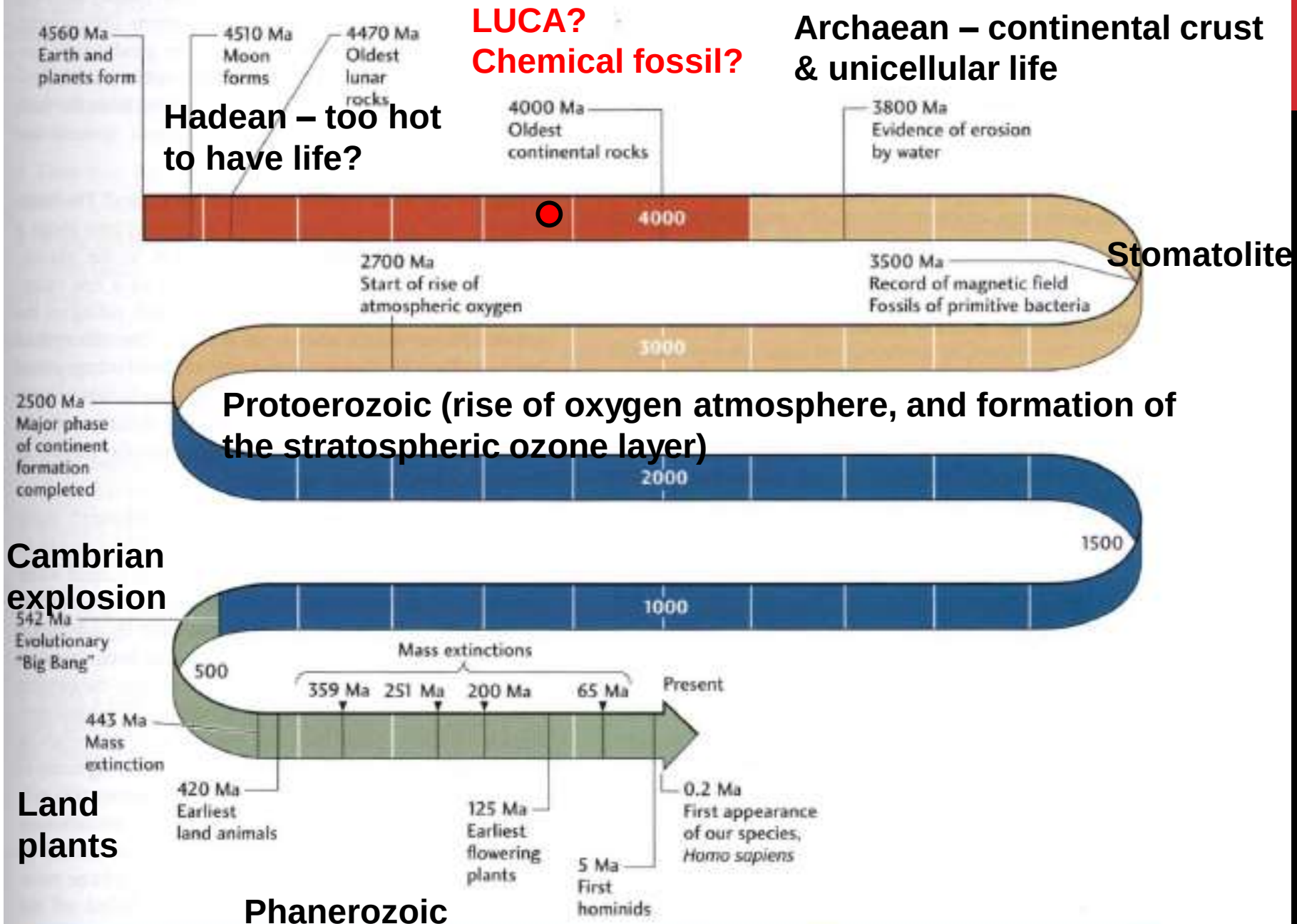
- The next smaller subdivision is the geological period.
- These periods form elements of a hierarchy of divisions into which geologists have split the Earth's history.
- A new period need not start with a mass extinction. It can start with the emergence of a new life form. Periods are often defined by so called index fossils (unique forms of life). Some boundaries between periods are marked by minor mass extinction events (e.g. Ordovician–Silurian extinction event or the Late Devonian extinction, near the Devonian–Carboniferous transition.)

EPOCH

- Epochs are most commonly used for the younger Cenozoic Era, where a greater collection of fossils has been found and paleontologists have more detailed knowledge of the events that occurred during those times.
- They are less commonly referred to for the other eras and eons, since less fossil evidence exists that allows us to form a clearer view of those time periods
- There are a lot of discussions regarding proper divisions of epochs particularly between marine and terrestrial geologists. Boundaries are not always very sharp.

Key Figure 1.13

The geologic record shows evidence of global geosystems operating in deep time.



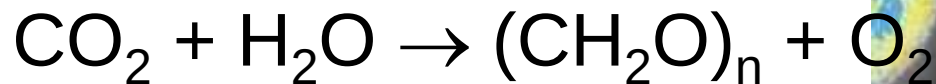
IMPORTANT MILESTONE IN THE ARCHEAN

Cyanobacteria:

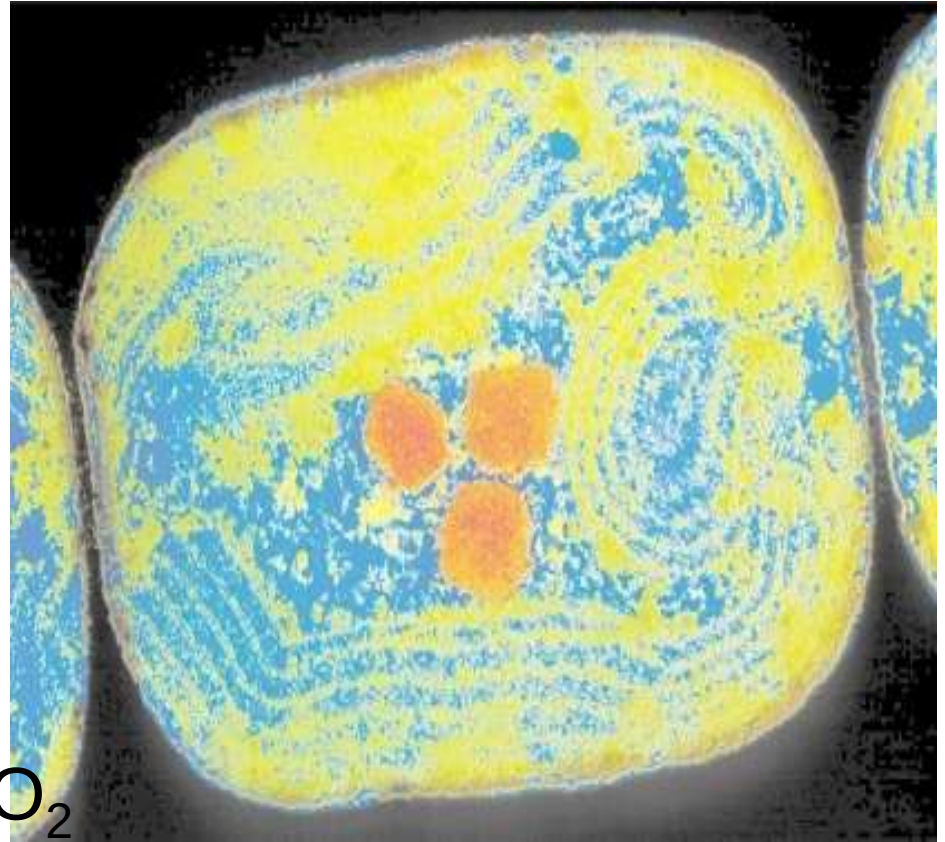
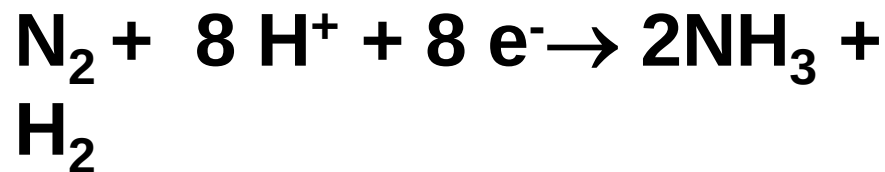
The first organisms making our planet habitable

Dated based on stromatolites to 3.5 Ga

Photosynthesis:



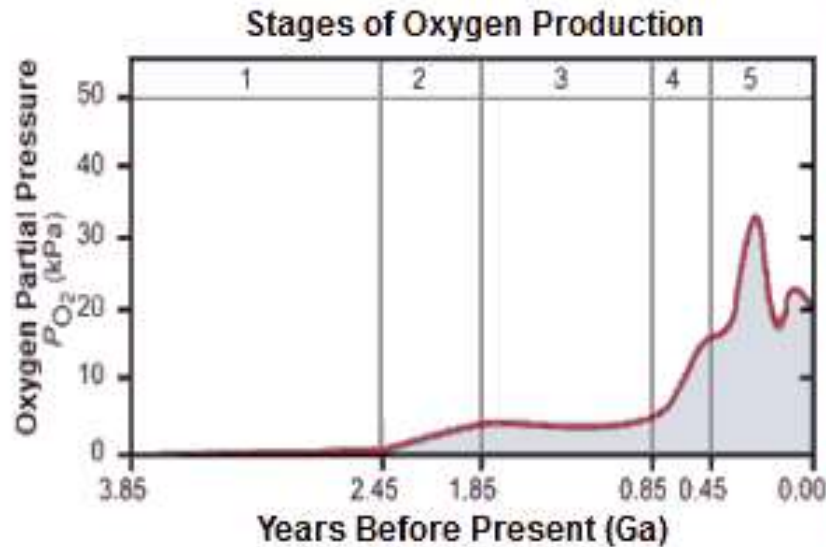
Nitrogen fixation:



3.5 Billion years ago some organisms developed the ability to convert CO_2 to organic carbon by photosynthesis, which also released O_2

They no longer depended on hydrothermal vents for their energy and set the stage for life leaving the ocean

PROTOEROZOIC THE RISE OF O₂



Banded Iron Formation (BIF) from Australia

Stromatolites are created by photosynthetic cyanobacteria, the oldest are 3.5 Ga old

The red bands are hematite an iron III mineral



It resulted in the extinction of many forms of chemoautotrophs to whom oxygen is toxic. Some chemoautotrophs still reside in extreme habitats e.g. around deep sea volcanoes as extremophiles.

THE FIRST MASS EXTINCTION WAS TRIGGERED BY OXYGEN

PALEOZOIC ERA

Cambrian Period



Almost all marine phyla evolved in this period.

Ordovician Period



Evolution of primitive fish, cephalopods, and coral

Silurian Period

Fully terrestrial life evolved, including early arachnids, fungi, and centipedes.

Devonian Period

On land, plant groups diversified incredibly in an event known as the Devonian Explosion when plants made lignin allowing taller growth and vascular tissue: the first trees evolved,

Carboniferous Period (359 to 299 million years ago)

Average global temperatures were high at the start and low at the end; the early Carboniferous averaged at about 20 degrees Celsius (but cooled to 10 °C during the Middle Carboniferous).

Tropical swamps dominated the Earth, and the lignin stiffened trees grew to greater heights and number. As the bacteria and fungi capable of eating the lignin had not yet evolved, their remains were left buried, which created much of the carbon that became the coal deposits of today

Permian Period

The Permian spanned from 299 to 252 million years ago and was the last period of the Paleozoic Era. At the beginning of this period, all continents joined together to form the supercontinent Pangaea

PERMIAN–TRIASSIC EXTINCTION EVENT

- Around 81% of all marine species and 70% of terrestrial vertebrate species became extinct. It was the largest known mass extinction of insects. Some 57% of all biological families and 83% of all genera became extinct.
- The eruption of the Siberian Traps coincide with the event. Evidence for widespread ocean acidification, **ocean anoxia (severe deficiency of oxygen)** and **euxinia (presence of hydrogen sulfide)** is found from the Late Permian to the Early Triassic. **CO₂ levels topped at 2000 ppm.**



MESOZOIC



70% of oil deposits existing today were formed in the Mesozoic age (252 to 66 million years ago), 20% were formed in the Cenozoic age (65 million years ago to present), and only 10% were formed in the Paleozoic age (541 to 252 million years ago).

The Mesozoic was a time of significant tectonic, climate, and evolutionary activity. The era witnessed the gradual rifting of the supercontinent Pangaea into separate landmasses that would move into their current positions during the next era. The climate of the Mesozoic was varied, alternating between warming and cooling periods. Overall, however, the Earth was hotter than it is today. Dinosaurs first appeared in the Mid-Triassic, and became the dominant terrestrial vertebrates in the Late Triassic or Early Jurassic, occupying this position for about 150 or 135 million years until their demise at the end of the Cretaceous. Birds first appeared in the Jurassic (however, true toothless birds appeared first in the Cretaceous), having evolved from a branch of theropod dinosaurs. The first mammals also appeared during the Mesozoic, but would remain small—less than 15 kg (33 lb)—until the Cenozoic. The flowering plants (angiosperms) appeared in the Early Cretaceous and would rapidly diversify throughout the period, replacing conifers and other gymnosperms as the dominant group of plants.

CRETACEOUS–PALEOGENE EXTINCTION EVENT



The Cretaceous–Paleogene (K–Pg) extinction event, also known as the Cretaceous–Tertiary (K–T) extinction, was a sudden mass extinction of three-quarters of the plant and animal species on Earth, approximately 66 million years ago. With the exception of some ectothermic species such as the sea turtles and crocodilians, no tetrapods weighing more than 25 kilograms (55 pounds) survived



In the geologic record, the K–Pg event is marked by a thin layer of sediment called the K–Pg boundary, which can be found throughout the world in marine and terrestrial rocks. **The boundary clay shows unusually high levels of the metal iridium, which is more common in asteroids than in the Earth's crust.**

The Deccan Trapps also erupted at the end of the Cretaceous



CENOZOIC

- The Cenozoic is also known as the Age of Mammals because the terrestrial animals that dominated both hemispheres were mammals – the Eutherians (placentals) in the northern hemisphere and the Metatherians (marsupials, now mainly restricted to Australia) in the southern hemisphere. The extinction of many groups allowed mammals and birds to greatly diversify so that large mammals and birds dominated the Earth. The continents also moved into their current positions during this era.

HOLOCENE EXTINCTION

- The Holocene extinction, otherwise referred to as the sixth mass extinction or Anthropocene extinction, is an ongoing extinction event of species during the present Holocene epoch (with the more recent time sometimes called Anthropocene) as a result of human activity.
- The included extinctions span numerous families of plants and animals, including mammals, birds, reptiles, amphibians, fishes and invertebrates.
- With widespread degradation of highly biodiverse habitats such as coral reefs and rainforests, as well as other areas, the vast majority of these extinctions are thought to be undocumented, as the species were undiscovered at the time of their extinction, or no one has yet discovered their extinction. The current rate of extinction of species is estimated at 100 to 1,000 times higher than natural background extinction rates.
- In *The Future of Life* (2002), Edward Osborne Wilson of Harvard calculated that, if the current rate of human disruption of the biosphere continues, one-half of Earth's higher lifeforms will be extinct by 2100.
- At present, the rate of extinction of species is estimated at 100 to 1,000 times higher than the background extinction rate, the historically typical rate of extinction (in terms of the natural evolution of the planet);
- The current rate of extinction is 10 to 100 times higher than in any of the previous mass extinctions in the history of Earth.

A NEW GEOLOGICAL ERA NORMALLY STARTS WITH A MASS EXTINCTION EVENT!

**THE ANTHROPOCENE MAY HAVE STARTED WITH THE
EXTINCTION OF THE ICE-AGE MEGAFUNA, WAS
FOLLOWED BY DEFORESTATION FOR AGRICULTURE
AND SMELTING. HOWEVER EXTINCTION BECAME
GLOBAL AND LARGE SCALE WITH MODERN
AGROCHEMICALS AND THE EXPANSION OF IRRIGATION
AGRICULTURE USING FOSSIL WATER DURING THE
GREAT ACCELERATION – BASED ON THIS FACT THE
ANTHROPOCENE COULD BECOME AN ERA**

ANTHROPOCENE

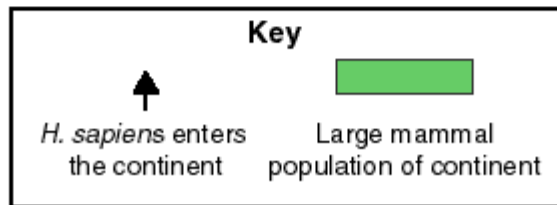
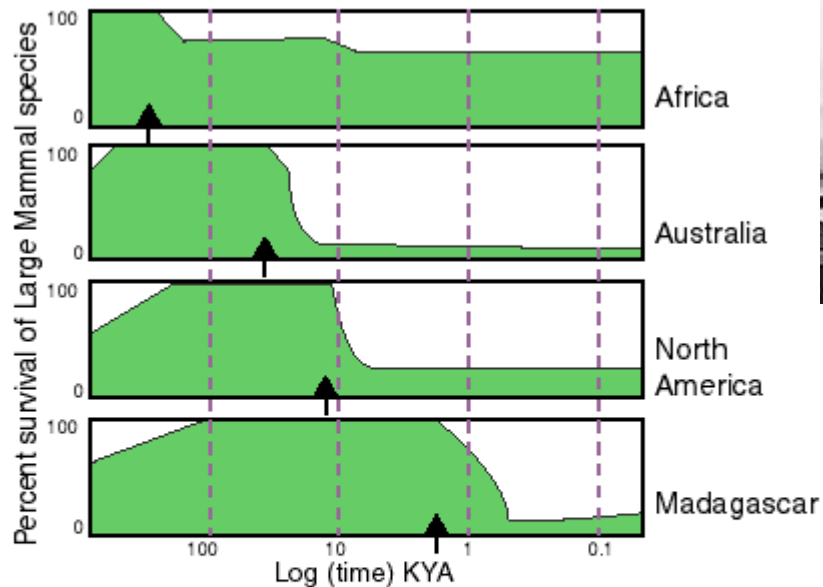
- The Anthropocene is **a proposed geological epoch** dating from the commencement of significant human impact on Earth's geology and ecosystems, including, but not limited to, anthropogenic climate change
- As of July 2020, neither the International Commission on Stratigraphy (ICS) nor the International Union of Geological Sciences (IUGS) has officially approved the term as a recognised subdivision of geologic time, although the Anthropocene Working Group (AWG) of the Subcommission on Quaternary Stratigraphy (SQS) of the ICS voted in April 2016 to proceed towards a formal golden spike (GSSP) proposal to define the Anthropocene epoch in the geologic time scale and presented the recommendation to the International Geological Congress in August 2016

Various start dates for the Anthropocene have been proposed, ranging from the beginning of the Agricultural Revolution 12,000–15,000 years ago, to as recent as the 1960s. The ratification process is still ongoing, and thus a date remains to be decided definitively, but **the peak in radionuclides fallout consequential to atomic bomb testing during the 1950s has been more favoured than others**, locating a possible beginning of the Anthropocene to the detonation of the first atomic bomb in 1945, or the Partial Nuclear Test Ban Treaty in 1963.

Geologists love sharp boundaries that can be dated (like the iridium layer of the C-Pg boundary) the nuclear test are hence more attractive because they will be traceable in every sediment rather than only sediments with large fossils –years into the future. Geologists are also conservative – they take a long time even to approve a new epoch.

MILESTONES OF MASS EXTINCTION

Megafauna extinction



Industrial revolution



The great Acceleration

Post WW2



The most silent extinction is that of insects and microbes

